Area 3: Mechanical Attachments

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

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3.A Bonds (Welding, Brazing, Adhesives)

1. Key concepts

- Fillet weld strength is set by the throat: t = 0.707h for an equal-leg fillet. Shear stress τ = F/(0.707hL).
- Weld groups under torsion or bending: treat the weld as a line. Find the unit properties J_u and I_u, then J = 0.707hJ_u. Add the primary (direct) and secondary (moment) shear as vectors.
- Weld symbols (AWS A2.4): a symbol below the reference line means arrow side; above means other side. A flag means field weld, a circle means all around, and the tail carries the process or spec.
- Adhesives are strong in shear and weak in peel. Lap joint: τ_avg = F/(bL).

2. Essential formulas

Name	Formula	Variables / units
Fillet throat stress	$\tau = F/(0.707hL)$	
Weld torsion	$\tau'' = Tr/J\ ;\ J = 0.707h \cdot J_u$	circle $J_u = 2\pi r^3$
Weld bending	$\tau = Mc/I\ ;\ I = 0.707h \cdot I_u$	circle $I_u = \pi r^3$
Adhesive lap	$\tau = F/(bL)$	

3. Handbook search keywords

fillet weld throat | weld group polar moment | welding symbols | adhesive lap joint

4. Worked examples

Example 1: Parallel fillet welds
Two 0.3125 in fillet welds, each 6 in long, carry 18,000 lb in shear. Find τ.

1. Throat area = 2(0.707)(0.3125)(6) = 2.651 in².
2. τ = 18,000/2.651 = 6,789 psi.

Answer: 6,789 psi

5. Common traps

- Using the leg h instead of the throat 0.707h.
- Adding the primary and secondary shear as scalars instead of vectors.
- Reversing arrow side and other side on the symbol.

3.A Bonds (Welding, Brazing, Adhesives): one-page cheat summary

- ► $t = 0.707h$
- ► $\tau = F/(0.707h\Sigma L)$
- ► $J = 0.707h \cdot J_u$
- ► Below line = arrow side

Formulas

Fillet throat stress	$\tau = F/(0.707hL)$
Weld torsion	$\tau'' = Tr/J$; $J = 0.707h \cdot J_u$
Weld bending	$\tau = Mc/I$; $I = 0.707h \cdot I_u$
Adhesive lap	$\tau = F/(bL)$

Search keywords

fillet weld throat, weld group polar moment, welding symbols, adhesive lap joint

Top traps

- Using the leg h instead of the throat 0.707h.
- Adding the primary and secondary shear as scalars instead of vectors.
- Reversing arrow side and other side on the symbol.

3.B Non-threaded Fasteners

1. Key concepts

- Pins, rivets and keys: check shear (single or double), bearing (crushing, σ_b = F/(dt)), and net-section tension of the plate, σ = F/((w − d)t).
- Keys: the force at the shaft surface is F = 2T/d. Shear τ = F/(wL); bearing on half the height σ = F/(L·h/2).
- Retaining rings, roll pins and shoulders locate parts axially.

2. Essential formulas

Name	Formula	Variables / units
Key force	$F = 2T/d$	
Key shear	$\tau = F/(wL)$	
Key bearing	$\sigma = 2F/(hL)$	square key h = w
Bearing on plate	$\sigma_b = F/(n \cdot d \cdot t)$	
Net section	$\sigma = F/((w - \Sigma d)t)$	

3. Handbook search keywords

key shear · rivet bearing stress · pin double shear · net section

4. Worked examples

Example 1: Square key length
A 0.375 in square key on a Ø1.5 in shaft carries 4,000 in-lb. Key Sy = 60 ksi, n = 2.5, Ssy = 0.577Sy. Find the minimum L.

1. F = 2(4,000)/1.5 = 5,333 lb.
2. Shear: L = Fn/(0.577·Sy·w) = 1.027 in.
3. Bearing: L = Fn/(Sy·w/2) = 1.185 in. Bearing governs.

Answer: 1.19 in

5. Common traps

- Using the diameter instead of the radius in F = T/r.
- Forgetting the bearing check on half the key height.

3.B Non-threaded Fasteners: one-page cheat summary

- ► $F = 2T/d$
- ► $\tau = F/wL$
- ► $\sigma_b = 2F/hL$
- ► $net = F/((w - d)t)$

Formulas

Key force	$F = 2T/d$
Key shear	$\tau = F/(wL)$
Key bearing	$\sigma = 2F/(hL)$
Bearing on plate	$\sigma_b = F/(n \cdot d \cdot t)$
Net section	$\sigma = F/((w - \Sigma d)t)$

Search keywords

key shear, rivet bearing stress, pin double shear, net section

Top traps

- Using the diameter instead of the radius in F = T/r.
- Forgetting the bearing check on half the key height.

3.C Threaded Fasteners

1. Key concepts

- Preload: F_i = 0.75A_tS_p for a reusable joint, 0.90A_tS_p for a permanent one. Torque T = KF_id with K ≈ 0.2 dry or 0.15 lubricated.
- Joint constant C = k_b/(k_b + k_m). The bolt picks up C·P of the external load and the members lose (1 − C)P. The joint separates at P_0 = F_i/(1 − C).
- Bolt fatigue: σ_a = CP/(2A_t) and σ_m = σ_a + F_i/A_t. Use Goodman with the preload line.
- Eccentric bolt groups: primary shear P/n plus secondary shear F = Mr/Σr², added as vectors.

2. Essential formulas

Name	Formula	Variables / units
Preload	$F_i = 0.75A_tS_p$ (reusable)	
Torque	$T = K \cdot F_i \cdot d$	
Joint constant	$C = k_b/(k_b + k_m)$	
Bolt load	$F_b = F_i + CP$	
Separation	$P_0 = F_i/(1 - C)$	
Eccentric group	$F'' = Mr_i/\Sigma r^2$	

3. Handbook search keywords

bolt preload · torque coefficient · joint stiffness constant · bolted joint separation · eccentric bolt group

4. Worked examples

Example 1: Preload, torque and separation
A 1/2-13 UNC grade 5 bolt (A_t = 0.1419 in², S_p = 85 ksi), reusable, K = 0.2, k_b = 3 and k_m = 10 (×10⁶ lb/in). Find F_i, T, the bolt load at P = 2,000 lb, and the separation load.

1. F_i = 0.75(0.1419)(85,000) = 9,046 lb.
2. T = 0.2(9,046)(0.5) = 905 in-lb.
3. C = 3/13 = 0.231; F_b = 9,046 + 0.231(2,000) = 9,508 lb.
4. P_0 = 9,046/(1 − 0.231) = 11,760 lb.

Answer: F_i ≈ 9,046 lb; T ≈ 905 in-lb; P_0 ≈ 11,760 lb

5. Common traps

- Giving the bolt the whole external load (it only takes C·P).
- Using the major-diameter area instead of A_t.
- Using 0.90 preload when the joint is reusable.

3.C Threaded Fasteners: one-page cheat summary

- ► $F_i = 0.75A_tS_p$
- ► $T = KF_id$
- ► $C = k_b/(k_b + k_m)$
- ► $F_b = F_i + CP$
- ► $P_0 = F_i/(1 - C)$

Formulas

Preload	$F_i = 0.75A_tS_p$ (reusable)
Torque	$T = K{\cdot}F_i{\cdot}d$
Joint constant	$C = k_b/(k_b + k_m)$
Bolt load	$F_b = F_i + CP$
Separation	$P_0 = F_i/(1 - C)$
Eccentric group	$F'' = Mr_i/\Sigma r^2$

Search keywords

bolt preload, torque coefficient, joint stiffness constant, bolted joint separation, eccentric bolt group

Top traps

- Giving the bolt the whole external load (it only takes C·P).
- Using the major-diameter area instead of A_t.
- Using 0.90 preload when the joint is reusable.